GECO and IECO: calibrated intensity ensembles from public real-time physical-model guidance

Project write-up · status as of 12 November 2026 · North Atlantic and East Pacific

Summary. The raw intensity forecasts of the physical ensembles, GEFS and the ECMWF ensemble (IFS-ENS), are far too weak and their members far too similar to be used as probabilistic guidance. GECO and IECO (GEFS / IFS-ENS Ensemble, Consensus-Oriented) are two twin products that turn them into calibrated intensity ensembles, using only data that are public in real time and available within the early-cycle window. Each member is NHC's corrected consensus HCCA plus that member's normalised scenario from the physical ensemble, scaled by a storm-specific predicted error. GECO uses the 30 GEFS member tracks in NHC's public files; IECO uses the 50 IFS-ENS member tracks that ECMWF publishes on its open-data mirror. Both run at every cycle from runs at least 6 h old, and neither needs any gridded data.

The fair yardstick is the best guidance already available, made probabilistic: "dressed HCCA". Over nine seasons (2018–2026) GECO beats it in every season, by 0.36 kt of CRPS on average (336 storms, every season significant); IECO beats it by 0.28 kt over 2018–2026 (240 storms, every season from 2020 significant). On 2025–26, with FNV3 (Google DeepMind) available, both are about 0.8 kt better than FNV3′s members at the same early timing, borderline significant over both seasons and significant in 2026, with a mean bias of about −1 kt against FNV3′s −6 kt. Their members evolve as smoothly as real storms. Rapid intensification remains the main weakness, as for every guidance system.

1 Motivation and design principles

AIFS-TC (Allen et al. 2026) showed that a cheap learned correction can lift an AI weather model's tropical-cyclone intensity forecasts to the operational frontier. This project asks whether the physical-model ensembles that forecasters already use can be turned, in the same spirit, into usable probabilistic intensity guidance. Their tracks are good, but at their resolution their intensities are 20–25 kt too weak on average and their members disagree far too little: in 2025–26 the raw ensembles' 10–90% ranges contain only 20–45% of outcomes.

Four principles shaped the design. Everything is built for the **early cycle**: an input may only be used if it is available within about three hours of the synoptic time, as for NHC's own consensus aids and the official forecast. Only **public data** are used. The products must run in **real time**. And every comparison uses NHC's verification conventions on **homogeneous** samples with storm-block bootstrap confidence intervals (Appendix A). FNV3 is a benchmark only and never an input. The target is the best guidance already available, so results are judged against HCCA and FNV3; GECO and IECO are two parts of one system, not competitors.

2 System

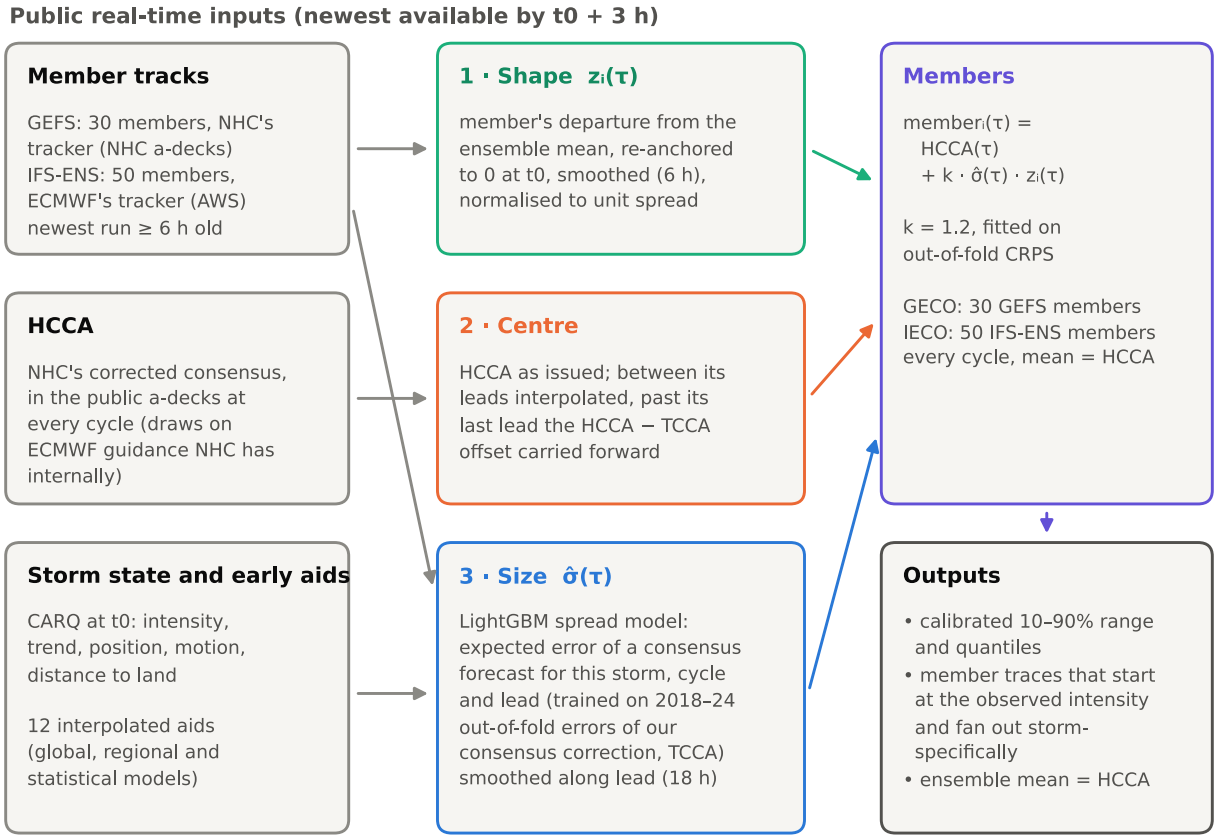


Figure 1: GECO and IECO. Each member takes its scenario shape from the physical ensemble (1), its centre from NHC's corrected consensus HCCA (2) and its spread from a storm-specific spread model (3).

Figure 1 shows the construction and Table 1 the inputs. All are public in real time.

Input	Source and use
GEFS member tracks (AP01–30)	NHC's tracker output in its public a-decks (`atcf/aid_public`), every cycle. GECO's shapes.
IFS-ENS member tracks (50)	ECMWF's own tracker output, BUFR files on the AWS open-data mirror (`s3://ecmwf-forecasts`), all four runs from January 2023; NHC's archived copies (EE01–50, identical values) before. IECO's shapes.
HCCA	NHC's corrected consensus, in the public a-decks at every cycle. The centre. It draws on ECMWF guidance that NHC has internally.
Storm state (CARQ), early aids	NHC's analysis at t_0 and twelve interpolated aids (GFS, GEFS mean, UKMET, Canadian, Navy, HWRF/HMON/HAFS, COAMPS-TC, DSHP, LGEM). Inputs to the spread model.
Consensus correction (TCCA)	Our LightGBM correction of the early-aid mean (no ECMWF, no SHIPS). Its quantile width and predicted 24-h change feed the spread model; it fills HCCA's gaps between and beyond its leads, and is GECO's fallback centre when HCCA is missing.

Table 1: Inputs. For both ensembles the newest run at least 6 h old is used, with lead times shifted by its age.

The member runs follow the early-cycle rule: the newest run that is at least 6 h old at the cycle time. NHC's GEFS tracks are in its files within that window. ECMWF posts its ensemble tracks 6.5–7.6 h after the start of the 00 and 12 UTC runs in 2025–26 (about 8.7 h in 2024) and about 6 h after the 06 and 18 UTC runs (AWS timestamps), so the 6-h-old run is available before every advisory, 9 h after its start, though with a thin margin in 2024. NHC's archive holds ECMWF member tracks mostly for the 00 and 12 UTC runs, so before 2023 IECO uses a 6-h-old run at half the cycles and a 12- or 18-h-old run at the others; from 2023 it is 6 h old at every cycle (79% of IECO's cases overall).

3 Construction

Shape from the physical ensemble. For member i at lead τ (every 12 h) the departure from the ensemble mean is re-anchored so that every member starts together at t_0, and smoothed along lead with a 6-h Gaussian kernel:

$$d_i(\tau) = [v_i(\tau) - \overline{v}(\tau)] - [v_i(0) - \overline{v}(0)], \quad \tilde{d}_i = \mathrm{smooth}(d_i), \quad z_i(\tau) = \tilde{d}_i(\tau) / \ \mathrm{sd}(\tau),$$

where $\mathrm{sd}(\tau)$ is the spread of the smoothed departures within that forecast, itself smoothed along lead (18 h). The z_i keep only the ensemble's scenario structure: which members intensify faster or slower, and when. The raw intensity level, about 20–25 kt too weak, is discarded. The 6-h kernel makes the members' step-to-step behaviour match the best track's (section 4.4) at no cost in skill.

Size from a spread model. A LightGBM regression predicts $\hat{\sigma}$, the expected absolute error of a consensus forecast for this storm, cycle and lead. Its features are the disagreement among the early aids, the spread of NHC's GEFS member tracks, the storm's intensity, recent trend, position and distance to land, the lead time, and the consensus correction's quantile width and predicted 24-h change; all 60 inputs are listed in Appendix B. It is trained on the absolute out-of-fold errors of the consensus correction from 2018–2024 (leave-one-season-out), smoothed along lead (18 h) and floored at 2 kt. On 2025–26 its prediction correlates with the actual error at 0.4 (0.33 within a fixed lead) and ranks forecasts well (Table 2).

Cases binned by predicted error	Cases	Predicted error (kt)	Actual error (kt)
Calmest fifth	1650	5.0	5.3
Second	1649	7.9	9.0
Middle	1649	11.0	13.6
Fourth	1649	14.6	15.6
Most volatile fifth	1650	21.0	19.0

Table 2: 2025–26 forecasts binned by predicted error $\hat{\sigma}$, against the consensus correction's actual mean absolute error.

Centre. HCCA as issued at its 12-hourly leads, without smoothing. Between HCCA's leads the HCCA-minus-TCCA offset is interpolated, and past its last lead the last offset is carried forward. Forecasts without HCCA (0.3% in 2025–26) fall back to our own consensus for GECO and are dropped for IECO.

Combine.

$$\mathrm{member}_i(\tau) = \mathrm{HCCA}(\tau) + k \cdot \hat{\sigma}(\tau) \cdot z_i(\tau),$$

clipped to 10–200 kt. Because the z_i have unit spread, the members' standard deviation is $k\hat{\sigma}$; k converts a predicted *mean absolute* error into a *standard deviation*. For normally distributed errors that factor is $\sqrt{\pi/2} \approx 1.25$. k is fitted, as $k(\tau) = c_0 + c_1\tau/100$, by minimising the fair continuous ranked probability score (CRPS) on out-of-fold 2018–2024 cases. The fit gives $c_1 = 0$ and $k = 1.2$ for both products, close to the normal-theory value, and 1.2–1.3 when refitted without any single season. The minimum is shallow: any k from 1.1 to 1.3 is within 0.02 kt of it (Appendix C). Because the z_i average to zero, the ensemble mean is HCCA: the products add a distribution, not a new mean.

This design separates *where* the ensemble is centred, *how wide* it is and *how its members differ*, and takes each from the source that does it best. It replaced earlier versions that centred the members on our own consensus correction (with or without gridded IFS-ENS features), which were level with dressed HCCA over 2018–2024 but fell behind it in 2026, and an IECO built from our own tracker on the gridded IFS-ENS fields, which could use only one run a day. Because the spread comes from the consensus error model and the raw ensembles contribute only their shapes, model upgrades that change a raw ensemble's spread (the ECMWF ensemble's roughly doubled between 2024 and 2025) do not change the products' spread.

4 Results

4.1 Yardsticks

Verification uses NHC's best track (2026: the working best track), leads 12–120 h, all four cycles and identical cases for every system in a comparison. For 2018–2024 every fitted ingredient is out of sample for the season scored: the consensus correction, the spread model and k are fitted without that season. 2025–26 use the production models, trained on 2018–2024. The primary yardstick is **dressed HCCA**: HCCA as a normal distribution whose width at each lead is HCCA's root-mean-square error in the other 2018–2024 seasons. Scoring HCCA as a single value would flatter any ensemble, whose CRPS is always below its own mean's absolute error. The second yardstick is **FNV3**, Google DeepMind's 50-member ensemble, taken from its newest run at least 6 h old, the same timing rule as ours; it is available for 2025–26.

4.2 Nine seasons against dressed HCCA

Season	GECO storms	GECO	HCCA dr. (G)	GECO − HCCA dr.	IECO storms	IECO	HCCA dr. (I)	IECO − HCCA dr.
2018	41	7.60	7.99	−0.39 [-0.54, -0.20]	1	9.97	10.32	−0.35 (1 storm)
2019	37	7.38	7.81	−0.43 [-0.59, -0.28]	1	6.06	6.17	−0.11 (1 storm)
2020	52	6.73	7.16	−0.43 [-0.60, -0.25]	44	6.91	7.24	−0.33 [-0.50, -0.17]
2021	40	6.60	6.97	−0.37 [-0.54, -0.23]	36	6.86	7.11	−0.26 [-0.45, -0.10]
2022	33	6.44	6.73	−0.28 [-0.50, -0.06]	31	6.64	6.92	−0.29 [-0.48, -0.07]
2023	39	6.58	6.85	−0.30 [-0.49, -0.11]	36	6.66	6.96	−0.33 [-0.49, -0.16]
2024	32	7.44	7.75	−0.31 [-0.49, -0.14]	30	7.82	7.97	−0.15 [-0.29, -0.03]
2025	30	7.77	8.08	−0.32 [-0.53, -0.13]	29	8.39	8.61	−0.22 [-0.44, -0.04]
2026	32	6.90	7.23	−0.33 [-0.45, -0.21]	32	7.16	7.47	−0.31 [-0.48, -0.17]
2018–24	274	6.97	7.33	−0.36 [-0.43, -0.29]	179	7.00	7.27	−0.28 [-0.36, -0.21]
2025–26	62	7.26	7.58	−0.32 [-0.43, -0.20]	61	7.65	7.92	−0.27 [-0.40, -0.15]
All	336	7.04	7.39	−0.36 [-0.41, -0.29]	240	7.19	7.46	−0.28 [-0.34, -0.21]

Table 3: Fair CRPS (kt) by season. Each product is scored on its own sample (GECO: every cycle with NHC's GEFS tracks and HCCA; IECO: every cycle with ECMWF member tracks and HCCA), each against dressed HCCA on the same cases, with 95% storm-bootstrap intervals. NHC's archive has ECMWF member tracks for only one storm in each of 2018 and 2019.

GECO beats dressed HCCA in all nine seasons, each significantly, by −0.36 kt [−0.41, −0.29] overall (Table 3). The gain is steady across the change from the 20-member GEFS to the 30-member GEFSv12 in September 2020 and the switch from HWRF to HAFS in 2023. IECO beats dressed HCCA in every season from 2020, each significantly, by −0.28 kt [−0.34, −0.21] overall, and by the same amount whether its run is 6 h old or older. Both are well calibrated: 76% (GECO) and 77% (IECO) of outcomes fall inside the 10–90% range, against 80% for a perfect ensemble, and IECO's spread is 0.91 of its error. On identical cases the two are within 0.04 kt of each other (0.04 kt [0, 0.07]). Since both means are HCCA, the gain comes entirely from the members' scenario shapes and the storm-specific spread, and the GEFS and IFS-ENS shapes turn out to be about equally informative.

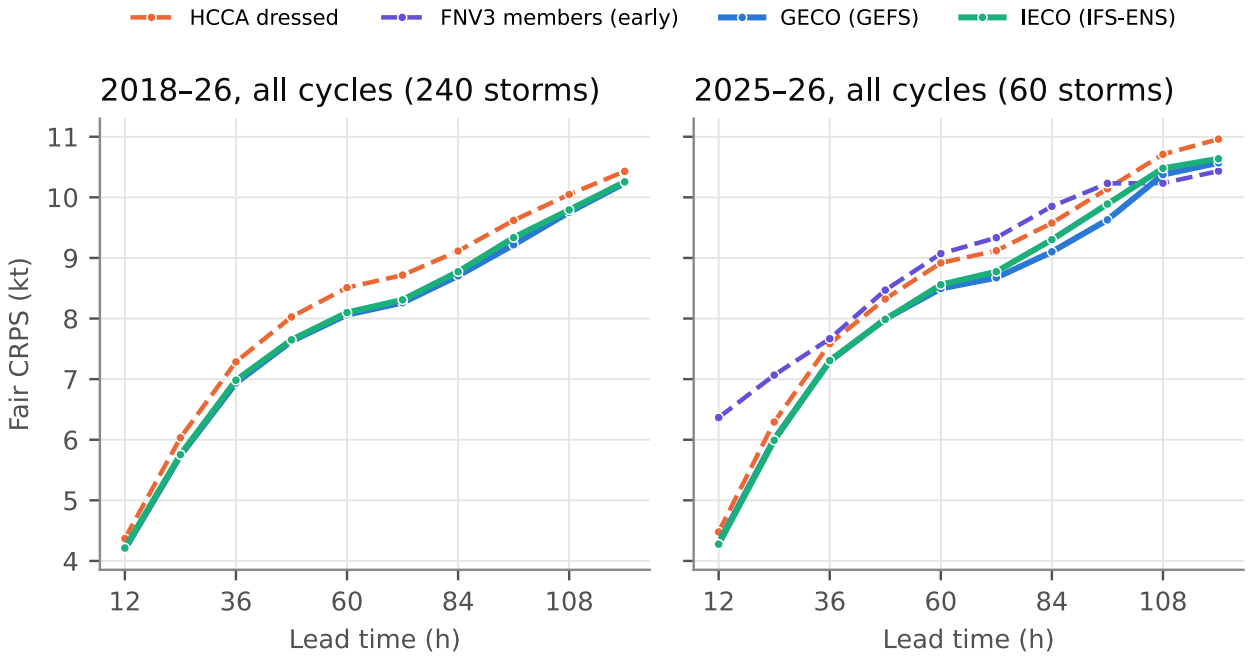


Figure 2: Fair CRPS by lead time on identical cases. Left: 2018–2026 (22015 cases where both products exist). Right: 2025–26 with FNV3 members (6345 cases).

The gain over dressed HCCA holds at every lead from 12 to 120 h (Figure 2).

4.3 2025–26 against FNV3

System	CRPS (kt)	Bias of mean (kt)	10–90% coverage	Spread / error	− HCCA dressed	− FNV3
GECO (GEFS)	7.59	−1.2	75%	0.87	−0.34 [-0.50, -0.21]	−0.80 [-1.65, +0.02]
IECO (IFS-ENS)	7.64	−1.3	75%	0.87	−0.29 [-0.42, -0.15]	−0.74 [-1.59, +0.07]
HCCA dressed	7.93	−1.2	83%	0.96		
FNV3 members (early)	8.39	−6.0	62%	0.69		
Raw GEFS members	22.64	−25.5	22%	0.20		
Raw IFS-ENS members	16.94	−20.7	45%	0.43		

Table 4: 2025–26, all cycles, leads 12–120 h, identical cases for every row (60 storms, 6345 cases). Spread / error is the mean member spread divided by the root-mean-square error of the ensemble mean (1 for a well-dispersed ensemble). Differences in CRPS with 95% storm-bootstrap intervals.

On the cases where all systems are available (Table 4), the CRPS difference from FNV3 is −0.8 kt [−1.65, 0.02] for GECO and −0.74 kt [−1.59, 0.07] for IECO: both intervals just reach zero over the two seasons. In 2026 both are significantly better (−1.05 kt [−1.85, −0.15] and −1.06 kt [−1.88, −0.17]); in 2025 the differences are smaller and uncertain. The interval for GECO excludes zero on slightly different samples, for example cycles at which FNV3's late run also exists (`docs/fnv3_compare.md`). Both products are better than FNV3 at every lead to 96 h, most clearly in the first two days, and slightly behind it at 108–120 h. FNV3's members are sharp but over-confident: their spread is 0.69 of their error, their mean is 6 kt too weak and only 62% of outcomes fall in their 10–90% range. GECO and IECO are nearly unbiased (−1.2 and −1.3 kt, as HCCA), and their spread follows their error at every lead, a little below it (0.87 overall; Figure 3). Their differences from dressed HCCA on these cases

are −0.34 kt [−0.5, −0.21] (GECO) and −0.29 kt [−0.42, −0.15] (IECO). The raw ensembles are two to three times worse than either.

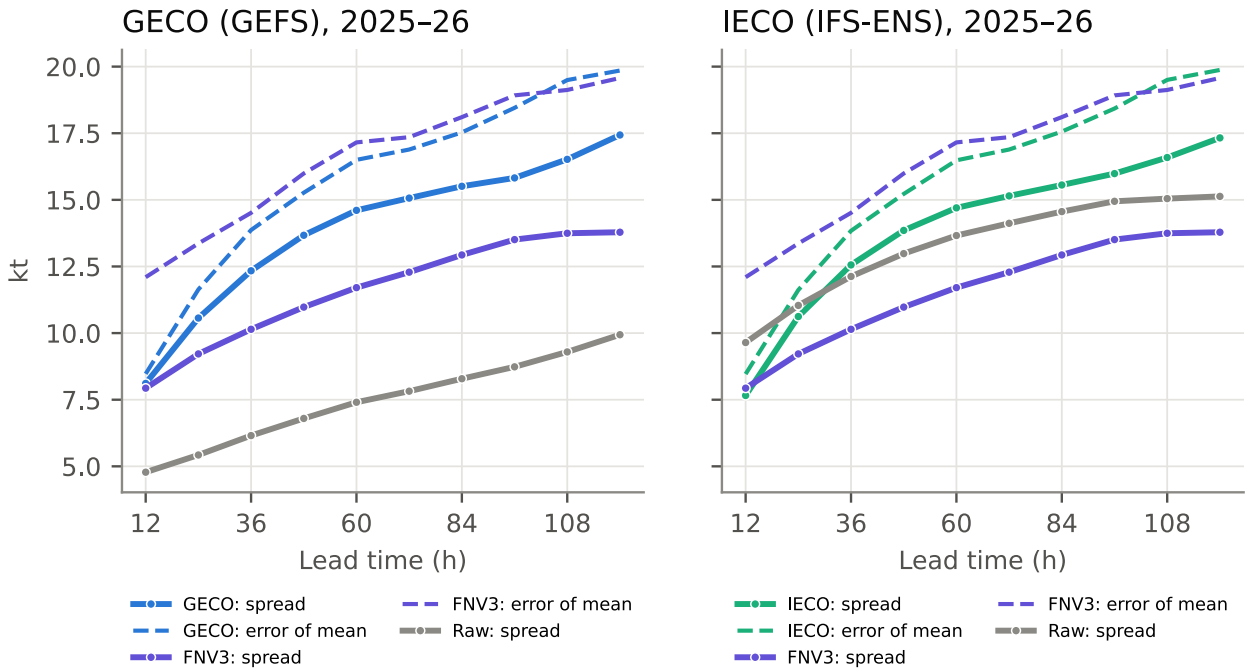


Figure 3: Mean member spread (solid) and the error of the ensemble mean (dashed) by lead time, 2025–26, same cases as Table 4. A well-calibrated ensemble has the two close together. The raw ensembles' errors (20–40 kt) are off the scale and only their spread is shown.

4.4 Physical consistency

Traces	Zig-zag rate	Mean |12-h change| (kt)	Share of 12-h changes > 30 kt
Best track	0.033	8.6	2.1%
FNV3 members	0.072	10.1	4.4%
Raw GEFS members	0.014	5.2	0.3%
Raw IFS-ENS members	0.037	7.0	1.1%
GECO (GEFS)	0.025	9.1	2.2%
IECO (IFS-ENS)	0.017	8.9	2.2%

Table 5: Step-to-step behaviour of intensity traces, 2025–26, leads 0–120 h. A zig-zag is two consecutive 12-h changes of opposite sign, each at least 10 kt.

Scores alone do not show whether individual members look like plausible storms, so we also measured how traces evolve from one time step to the next (Table 5). GECO and IECO members change by about as much per 12 h as the best track, exceed 30 kt in 12 h about as often, and reverse direction less often. FNV3's members zig-zag about twice as often as the best track at 12-h spacing, and at their native 6-h spacing they reverse direction 21 times as often, which is the short-term jitter visible in their traces.

4.5 Case studies

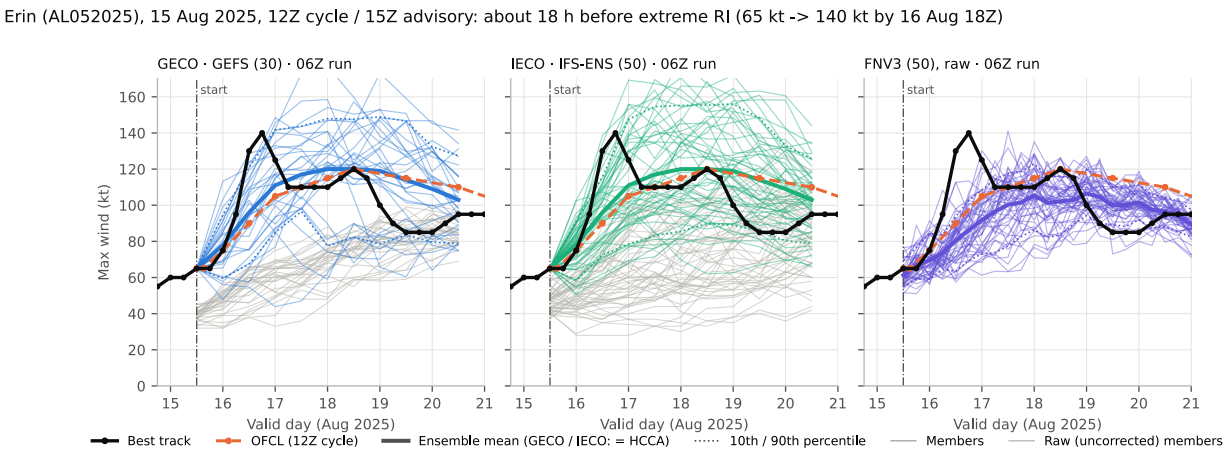


Figure 4: Hurricane Erin, 15 August 2025, 12 UTC cycle, about 18 h before its extreme rapid intensification (65 kt to 140 kt by 16 August 18 UTC). GECO (NHC's 30 GEFS member tracks) and IECO (ECMWF's 50 member tracks) from the 06 UTC runs, FNV3′s 06 UTC members, the best track and the official forecast of the same cycle. In grey, the raw members, starting at the models' own intensity.

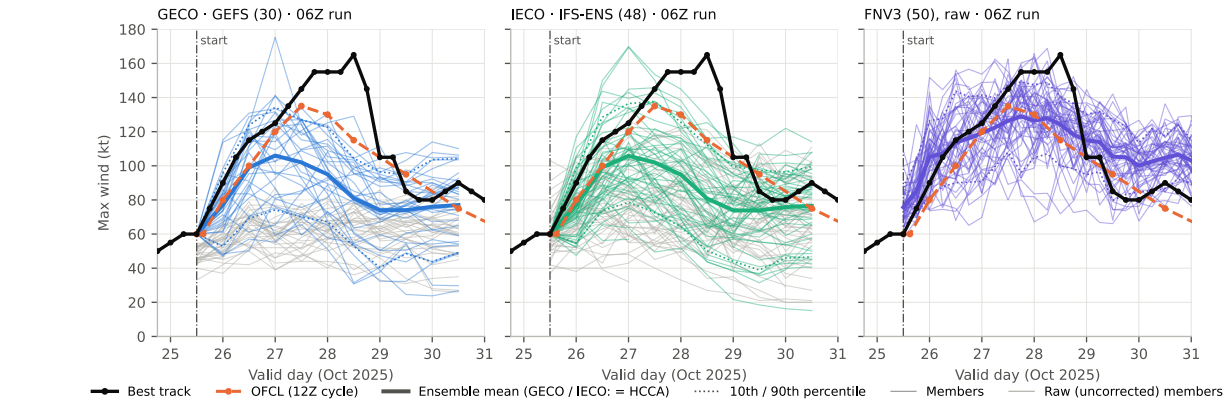


Figure 5: Hurricane Melissa, 25 October 2025, 12 UTC cycle, at the start of its rapid intensification to 165 kt.

The cases show what the numbers mean in practice. Every panel is for the same cycle, as the products would run in real time. For Erin (Figure 4) the spread model recognises a volatile situation: both ensembles fan out widely in the first day and their upper members reach the 140-kt peak, whereas FNV3′s members stay mostly below about 125 kt and the official forecast peaks near 120 kt. For Melissa (Figure 5) both ensembles rise quickly from the observed intensity and fan out; their means, which are HCCA, peak well below the 165-kt best track, but the upper members reach 140–170 kt, so the extreme outcome lies within the ensemble. FNV3′s mean comes closer to the peak. Eight further cases are in Appendix D. A larger set of 39 cases, including eleven high-profile storms from 2018–2024 (Michael, Florence, Dorian, Laura, Iota, Ida, Ian, Lee, Otis, Beryl, Milton) scored out of sample, is in `docs/case_plots.pdf`.

5 Real-time operation

At each cycle the products need NHC's public a-decks (CARQ, the early aids, HCCA and the GEFS member tracks) and ECMWF's newest ensemble track file at least 6 h old. All are small text or BUFR files of a few MB at most. The computation (shapes, the spread model, combination) takes seconds on one core, and no gridded data are read. The inputs are therefore all available in time, but two pieces are not yet in place: the arrival times

of HCCA and the GEFS member tracks relative to the forecast deadline have not been measured (a poller is ready for the next active storm), and the trained models are not yet saved for a stand-alone real-time runner.

6 Limitations and next steps

The products are symmetric around HCCA, which, like every guidance system, under-predicts rapid intensification. In such situations the truth still falls above the 90th percentile more often than it should, and some lower members describe weakening that does not happen. Making the spread asymmetric did not help, because which borderline situations will intensify rapidly cannot be identified well enough in advance. Both products inherit HCCA's errors in the mean. The spread model predicts our consensus correction's errors rather than HCCA's, and k absorbs the difference. The test seasons contain about 60 storms, so only large differences are significant, and the 2026 verification uses NHC's working best tracks. Before 2020, NHC's archive holds ECMWF member tracks for only two storms.

The next steps, in order:

- A real-time runner: persist the trained models, poll NHC's public files and ECMWF's track files, and write ATCF-format members.
- Measure the arrival times of HCCA, the GEFS member tracks and ECMWF's track files during the next active storms.
- Re-score 2026 once NHC finalises its best tracks.
- A convolutional network on multi-level gridded fields, combined with this construction, when compute allows; the present products use no gridded data at all.

Appendix A · Verification framework

All verification follows NHC's conventions. Truth is the final best track (b-deck; 2026: the working best track); a case counts only if the system is a tropical or subtropical cyclone at both the initial and the valid time; and every comparison is homogeneous. Ensembles are scored by the fair continuous ranked probability score (CRPS), the coverage of the 10–90% range and the ratio of spread to the error of the mean. Confidence intervals resample whole storms (block bootstrap), as in the AIFS-TC paper, because cases from the same storm are strongly correlated.

As a check, the harness recomputes NHC's published 2024 Atlantic verification. The official-forecast table is reproduced with identical case counts and errors within 0.1 kt, and the homogeneous early-model table with identical case counts and 94 of 96 errors within 0.2 kt (Figure 6). The AIFS-TC paper's 2025 values for OFCL and FNV3 are reproduced within their confidence intervals.

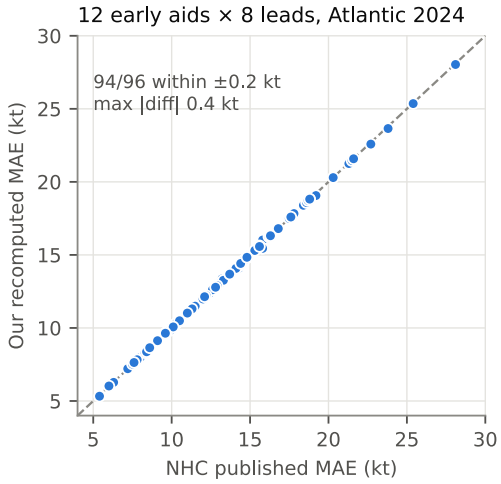


Figure 6: Recomputed versus published errors for NHC's 2024 homogeneous early-aid verification; each point is one aid at one lead time.

Appendix B · Spread-model inputs

The spread model is a LightGBM regression (500 trees of 15 leaves, learning rate 0.03, at least 200 cases per leaf, 70% of features and 80% of cases per tree, L2 penalty 1) whose target is the absolute error of the consensus correction, $|v_{\mathrm{obs}} - \mathrm{TCCA}|$, at each storm, cycle and lead (12–168 h). Table 6 lists every input. All are available by t_0 + 3 h; missing values (an aid not run, a lead beyond an aid's range) are left missing and handled by the trees. Early aids are NHC's interpolated versions, with the 6-h and 12-h versions merged. There are no ECMWF, SHIPS or gridded inputs. The spread model never sees the IFS-ENS members: IECO uses the same $\hat{\sigma}$ as GECO.

Group	Variable	Description
Storm state (CARQ, t0)	vmax0	maximum wind at t0 (kt)
Storm state (CARQ, t0)	pmin0	minimum pressure at t0 (hPa)
Storm state (CARQ, t0)	lat0	latitude at t0
Storm state (CARQ, t0)	lon0	longitude at t0
Storm state (CARQ, t0)	dv6	intensity change over the previous 6 h
Storm state (CARQ, t0)	dv12	intensity change over the previous 12 h
Storm state (CARQ, t0)	dv24	intensity change over the previous 24 h
Storm state (CARQ, t0)	u_motion	eastward motion over the previous 12 h (km/h)
Storm state (CARQ, t0)	v_motion	northward motion over the previous 12 h (km/h)
Storm state (CARQ, t0)	dist2land0	distance to land at t0 (km)
Storm state (CARQ, t0)	basin_ep	East Pacific (1) or Atlantic (0)

Group	Variable	Description
Storm state (CARQ, t0)	doy	day of the year of t0
Lead time	lead	lead time τ (h)
Early-aid consensus	n_aids	number of early aids available at this lead
Early-aid consensus	ref_vmax	equal-weight mean intensity of the available early aids (the reference)
Early-aid consensus	ref_pmin	reference pressure: CARQ pressure plus the GEFS members' mean pressure change
Early-aid consensus	aid_vmax_std	standard deviation of the early aids' intensities
Early-aid consensus	aid_vmax_range	range (max − min) of the early aids' intensities
Early-aid consensus	lat_c	mean forecast latitude of the early aids
Early-aid consensus	lon_c	mean forecast longitude of the early aids
Early-aid consensus	dist2land_c	distance to land at the mean forecast position (km)
GEFS members (NHC tracks)	gefs_vmax_mean	mean member intensity at this lead
GEFS members (NHC tracks)	gefs_vmax_std	standard deviation of member intensities
GEFS members (NHC tracks)	gefs_vmax_p10	10th percentile of member intensities
GEFS members (NHC tracks)	gefs_vmax_p90	90th percentile of member intensities
GEFS members (NHC tracks)	gefs_n	number of members tracked at this lead
GEFS members (NHC tracks)	gefs_mslp_mean	mean member minimum pressure at this lead
GEFS members (NHC tracks)	gefs_vmax0_mean	mean member intensity valid at t0
GEFS members (NHC tracks)	gefs_mslp0_mean	mean member pressure valid at t0
GEFS members (NHC tracks)	gefs_ri_frac	share of members rising ≥ 30 kt in the 24 h ending at this lead
GEFS members (NHC tracks)	gefs_age	age of the GEFS run at t0 (6 or 12 h)
GEFS members (NHC tracks)	gefs_offset0	mean member intensity at t0 minus CARQ intensity
Consensus correction (TCCA)	tcca_mean	corrected mean intensity at this lead (out of fold)
Consensus correction (TCCA)	tcca_qwidth	width of its 10–90% quantile range
Consensus correction (TCCA)	tcca_dv24	its predicted intensity change over the 24 h ending at this lead
Consensus correction (TCCA)	tcca_minus_v0	its predicted intensity minus the CARQ intensity at t0
Early aids (interpolated)	AVNI_vmax, AVNI_dv12	GFS: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	AEMI_vmax, AEMI_dv12	GEFS mean: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	EGRI_vmax, EGRI_dv12	UKMET: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	CMCI_vmax, CMCI_dv12	Canadian (CMC): intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	NVGI_vmax, NVGI_dv12	Navy (NAVGEM): intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	HWFI_vmax, HWFI_dv12	HWRF: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	HMNI_vmax, HMNI_dv12	HMON: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	HFAI_vmax, HFAI_dv12	HAFS-A: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	HFBI_vmax, HFBI_dv12	HAFS-B: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	CTCI_vmax, CTCI_dv12	COAMPS-TC: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	DSHP_vmax, DSHP_dv12	SHIPS with decay over land: intensity at this lead; its change over the preceding 12 h
Early aids (interpolated)	LGEM_vmax, LGEM_dv12	LGEM: intensity at this lead; its change over the preceding 12 h

Table 6: Every input of the spread model (60 variables; each early aid contributes two), one row per storm, cycle t_0 and lead time.

Appendix C · Choice of the spread factor k

k is chosen by minimising the out-of-fold CRPS. Table 7 shows the CRPS, the coverage of the 10–90% range and the spread / error ratio of both products over 2018–2024 for constant k (leads 12–120 h; 26672 GECO and 15491 IECO cases). Three points explain the choice of 1.2:

- **It is the CRPS minimum**, for both products. A narrower ensemble (smaller k) misses too many outcomes; a wider one is penalised for spreading probability where outcomes rarely fall. The minimum is shallow between 1.1 and 1.3 (within 0.02 kt).
- **It matches normal theory.** The members' standard deviation is $k\hat{\sigma}$ and $\hat{\sigma}$ predicts a mean absolute error, so for normally distributed errors the calibrated factor would be $\sqrt{\pi/2} \approx 1.25$.
- **It is slightly sharper than a spread that matches the error.** The errors are heavier-tailed than normal (their root-mean-square is 1.41 times their mean absolute value, against 1.25 for a normal distribution), so spread / error = 1 needs $k \approx 1.3$–1.4. The CRPS prefers a slightly narrower ensemble, because $\hat{\sigma}$ ranks forecasts only imperfectly: the extra width would be spent mostly on cases that do not need it. The cost is coverage a little below 80% (76–78% at $k = 1.2$).

The fit allowed k to vary linearly with lead time (c_1 from −0.4 to +0.4 in steps of 0.1; c_0 from 0.6 to 2.0 in steps of 0.1); the best slope was zero, so k is constant. Refitting without each 2018–2024 season in turn gave 1.2 or 1.3 with slopes of 0 to −0.1.

k	GECO CRPS	GECO coverage	GECO spread/ error	IECO CRPS	IECO coverage	IECO spread/ error
0.60	8.018	46%	0.45	7.516	47%	0.46
0.80	7.651	58%	0.59	7.195	59%	0.62
1.00	7.454	68%	0.74	7.031	70%	0.77
1.10	7.406	72%	0.82	6.996	74%	0.85
1.20	7.387	76%	0.89	6.987	78%	0.93
1.25	7.388	78%	0.92	6.992	79%	0.96
1.30	7.394	79%	0.96	7.001	81%	1.00
1.40	7.422	82%	1.03	7.035	83%	1.08
1.60	7.535	86%	1.18	7.153	87%	1.23
2.00	7.924	92%	1.45	7.539	93%	1.52

Table 7: Out-of-fold 2018–2024 fair CRPS (kt), 10–90% coverage (ideal 80%) and spread / error (1 for a well-dispersed ensemble) for constant k. GECO and IECO are on their own cases, so their CRPS values are not directly comparable.

Appendix D · Additional case studies

All panels are for the 12 UTC cycle: GECO and IECO from the 06 UTC runs, FNV3 from its 06 UTC run.

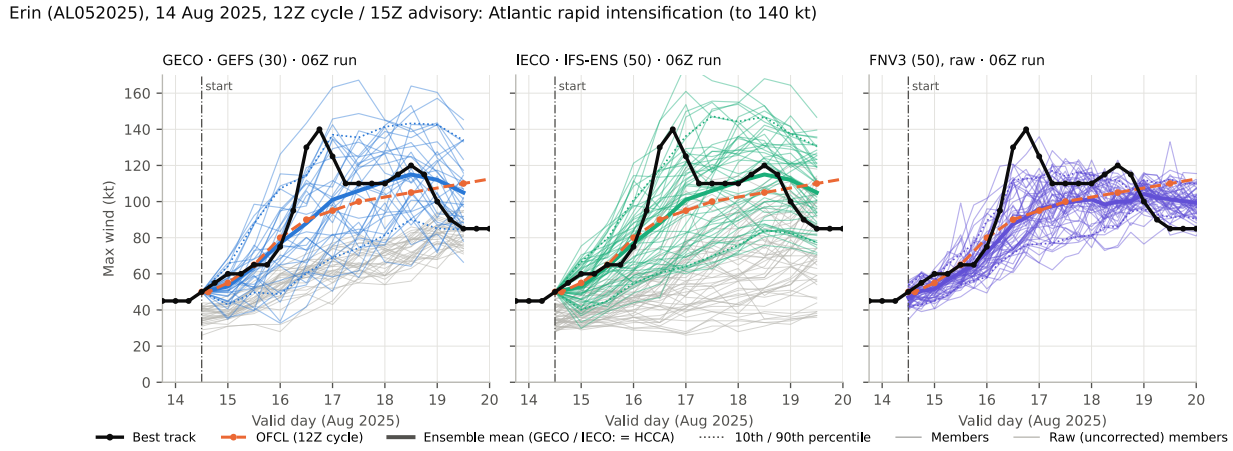


Figure 7: Hurricane Erin, 14 August 2025, two days before its rapid intensification.

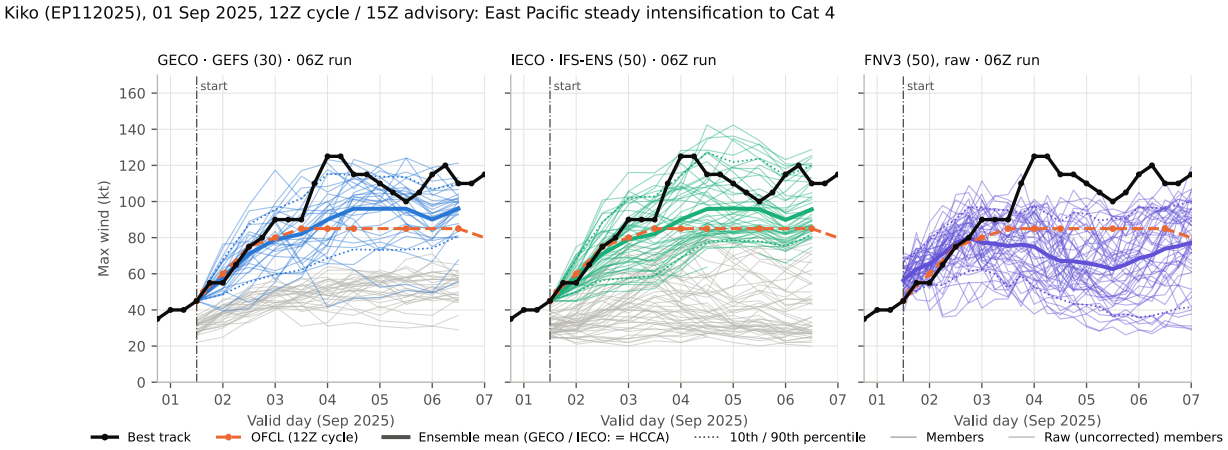


Figure 8: Hurricane Kiko, 1 September 2025, East Pacific: steady intensification to a category 4 hurricane.

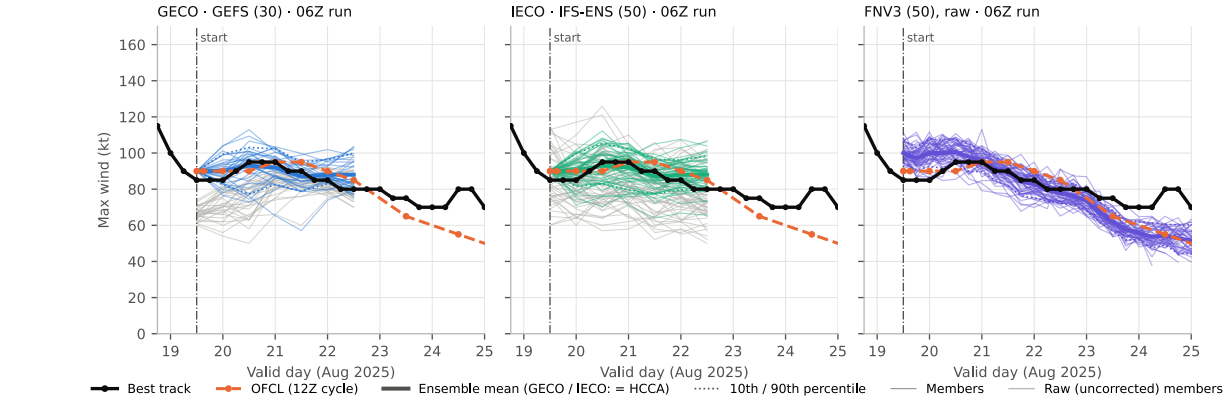


Figure 9: Hurricane Erin, 19 August 2025: recurvature and extratropical transition.

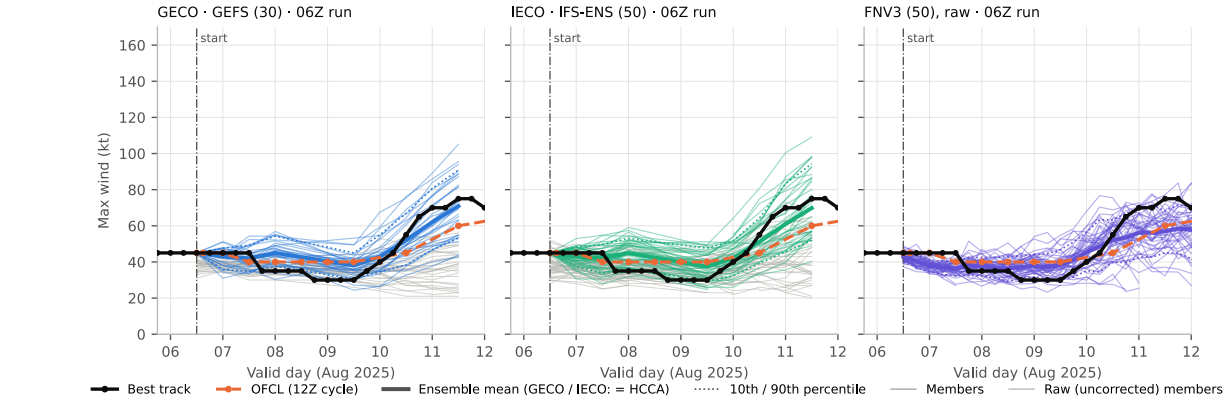


Figure 10: Tropical Storm Henriette, 6 August 2025: a weak, long-lived storm.

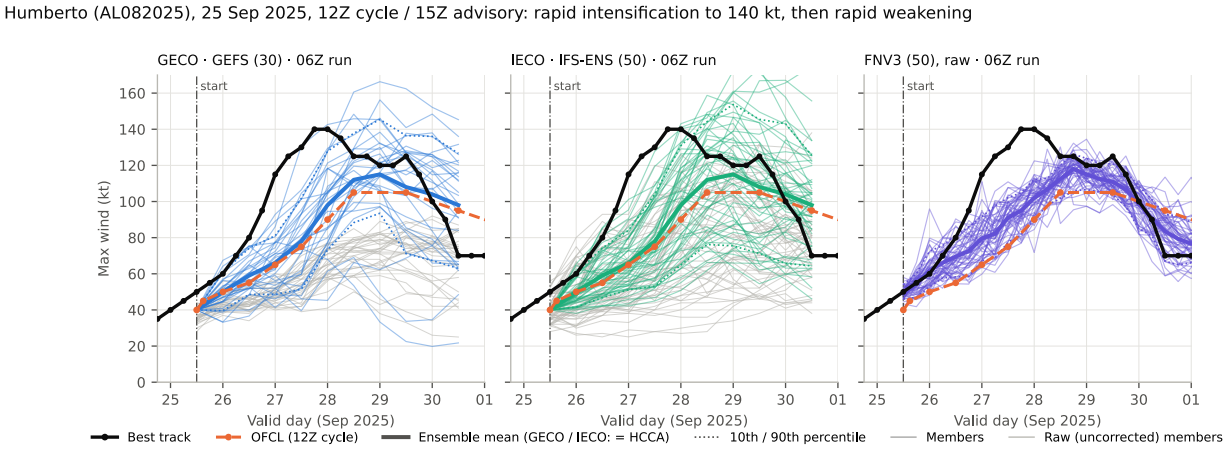


Figure 11: Hurricane Humberto, 25 September 2025: rapid intensification to 140 kt within three days, then rapid weakening.

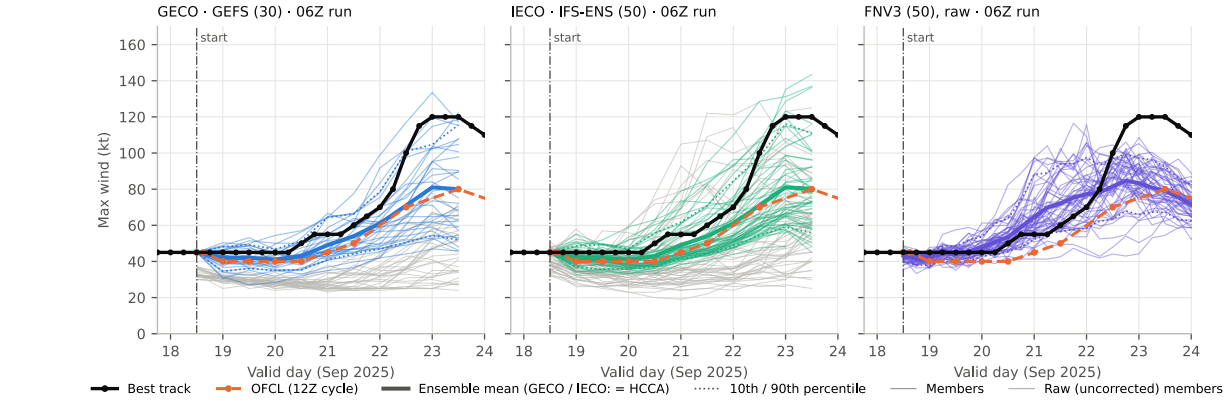


Figure 12: Hurricane Gabrielle, 18 September 2025: weak for four days, then rapid intensification to 120 kt.

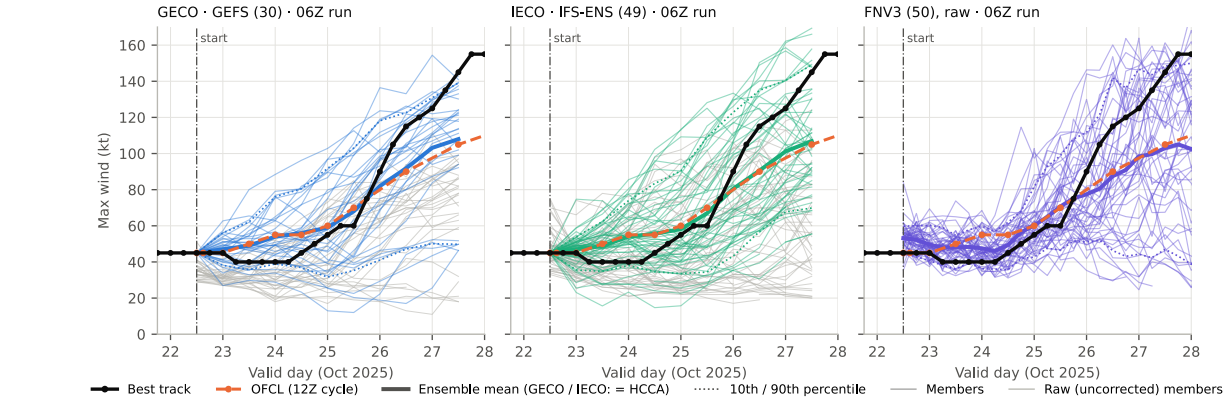


Figure 13: Hurricane Melissa, 22 October 2025, three to four days before its rapid intensification.

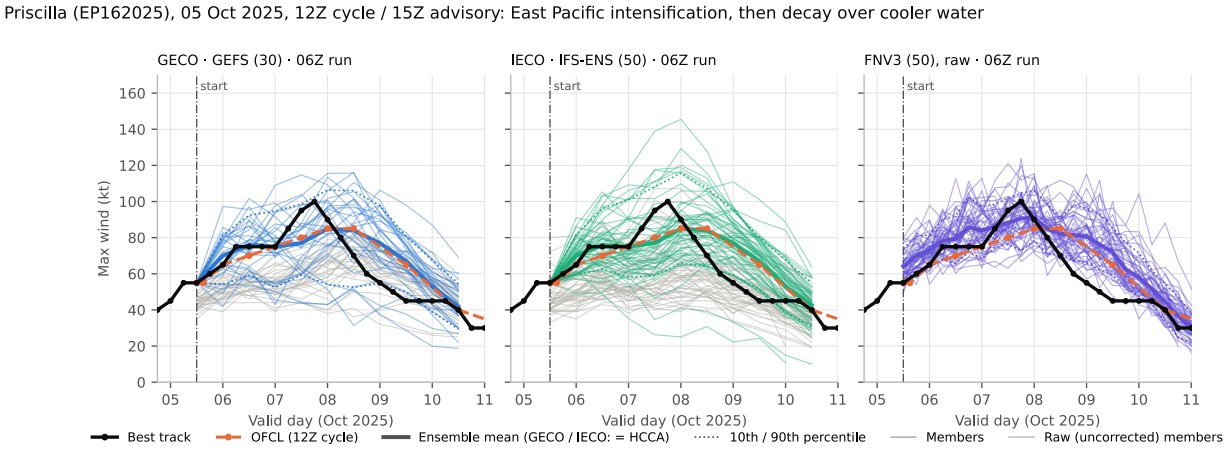


Figure 14: Hurricane Priscilla, 5 October 2025, East Pacific: rise to 100 kt and decay over cooler water.

Appendix E · Glossary and reproduction

CARQ NHC's operational analysis of position and intensity at t_0, with the preceding 24 h.
OFCL the official NHC forecast. HCCA: NHC's corrected consensus.
Early / late available / not available in time for the forecast cycle that starts at the synoptic time.
FNV3 Google DeepMind's cyclone model (Weather Lab archive, 50 members).
CRPS continuous ranked probability score, the probabilistic analogue of the mean absolute error (lower is better).
Dressed HCCA HCCA as a normal distribution with its out-of-season error by lead.
TCCA our consensus correction of the early-aid mean (no ECMWF, no SHIPS).
GECO GEFS Ensemble, Consensus-Oriented: NHC's 30 GEFS member tracks around HCCA, every cycle.
IECO IFS-ENS Ensemble, Consensus-Oriented: ECMWF's 50 ensemble member tracks around HCCA, every cycle.

Reproduce. The whole chain, resumable: `scripts/rerun_chain.sh start`. Inputs: `python -m tcic.data.atcf` (NHC decks), `scripts/fetch_ecmwf_tracks.py` and `python -m tcic.data.ecmwf_tf` (ECMWF tracks). Consensus correction and spread model: `python -m tcic.consensus.features`, `scripts/m15_run.py`, `python -m tcic.ensemble.members`, `scripts/ens_run.py`, `scripts/ens_calib.py`. Products: `src/tcic/ensemble/geco.py`; GECO `scripts/ens_geco.py` and `scripts/ens_multiseason.py`; IECO `scripts/ens_ieco.py`. Comparisons and figures: `scripts/fnv3_compare.py`, `scripts/report_figures_ens.py`, `scripts/case_plots.py`. Detailed results: `docs/ensemble_multiseason.md`, `docs/ensemble_ieco.md`, `docs/fnv3_compare.md`; history in `plan.md`.